PCBC H Parametric & Sequences Test (Practice A) **Name**_____ _____/16

SHOW ALL WORK FOR FULL CREDIT. ***Non-Calculator*** **You have the entire period.**

Honor Code (copy below & sign): On my honor, before God, I have neither given, nor received any aid on this test.

_____

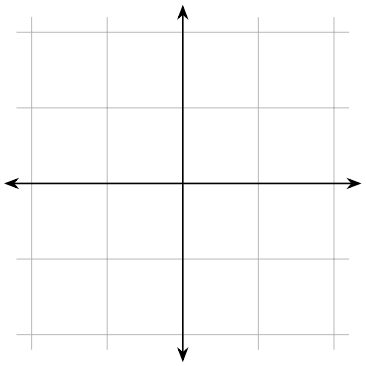
_____ _____

Signature

Write the pair of parametric equations in rectangular form, including the domain. Then sketch the curve, including direction. (3 points each)

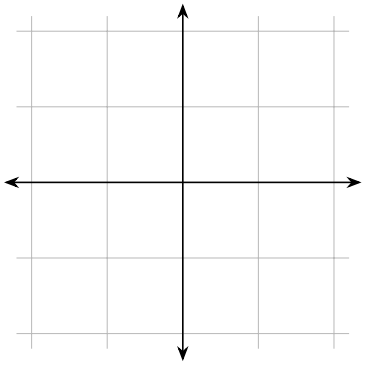
1. $x = 2t + 2, \quad y = t^2 + 2t + 1, \quad -3 \leq t \leq 1$

SIMPLIFIED rectangular form: _____



2. $x = \sqrt{t}, \quad y = 2t - 4, \quad 0 \leq t \leq 4$

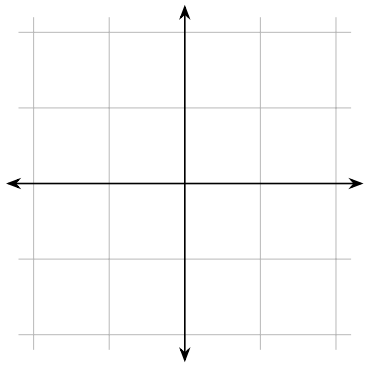
SIMPLIFIED rectangular form: _____



Write the pair of parametric equations in rectangular form. Then sketch the curve. (3 points each)

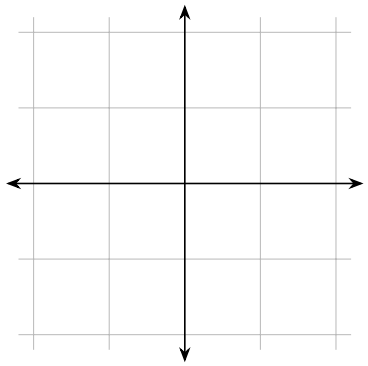
3. $x = 3\cos t, \quad y = 5\sin t, \quad 0 \le t \le 2\pi$

rectangular form: _____



4. $x = 2\sec t, \quad y = 3\tan t$

rectangular form: _____



Convert the rectangular equation to parametric form. Give two different sets of answers. (2 points)

5. $y = -3(x+2)^2 + 5$

The given pattern continues. Write down the nth term of the sequence $\{a_n\}$ suggested by the pattern. (2 points)

6. $\frac{1}{3}, -\frac{2}{9}, \frac{3}{27}, -\frac{4}{81}, \ldots$

Answer Key

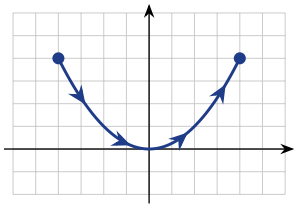
Work on the left, the reason beside it. Boxed = final answer.

Day 1: Non-Calculator

1. $x = 2t+2,\ y = t^2+2t+1,\ -3 \le t \le 1$

t	x	y
-3	-4	4
-2	-2	1
-1	0	0
0	2	1
1	4	4

brackets: plug in both endpoints



solid dots, no end arrows: closed interval

$x-2 = 2t \ \Rightarrow\ t = \frac{x-2}{2}$ — *isolate t in x: one answer, no $\pm$*

$y = \left(\frac{x-2}{2}\right)^2 + 2\left(\frac{x-2}{2}\right) + 1$ — *substitute*

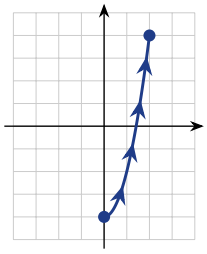
$y = \frac{x^2-4x+4}{4} + x - 2 + 1$

$\boxed{y = \frac{1}{4}x^2 \ \text{ for } x \text{ in } [-4,4]}$ — *SIMPLIFIED = y alone; domain = x from the t-chart*

Other way: $y = (t+1)^2$ and $t+1 = \frac{x}{2}$, so $y = \frac{x^2}{4}$.

2. $x = \sqrt{t},\ y = 2t-4,\ 0 \le t \le 4$

t	x	y
0	0	-4
1	1	-2
4	2	4



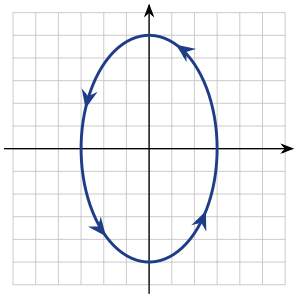
$x^2 = t$ — *square both sides: one answer for t*

$y = 2x^2 - 4$ — *substitute*

$\boxed{y = 2x^2 - 4 \ \text{ for } x \text{ in } [0,2]}$ — *$x = \sqrt{t}$ is never negative, so only the right half*

3. $x = 3\cos t,\ y = 5\sin t,\ 0 \le t \le 2\pi$

t	(x,y)
0	$(3,0)$
$\frac{\pi}{2}$	$(0,5)$
π	$(-3,0)$
$\frac{3\pi}{2}$	$(0,-5)$
2π	$(3,0)$



counterclockwise from $(3,0)$

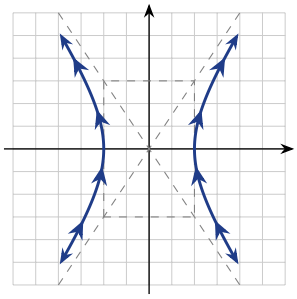
$\cos t = \frac{x}{3}, \quad \sin t = \frac{y}{5}$ — *solve each for the trig function*

$\sin^2 t + \cos^2 t = 1 \Rightarrow \left(\frac{y}{5}\right)^2 + \left(\frac{x}{3}\right)^2 = 1$ — *identity*

$\boxed{\frac{x^2}{9} + \frac{y^2}{25} = 1}$ — *whole ellipse is traced: no restriction*

4. $x = 2\sec t,\ y = 3\tan t$ (no interval given)

t	(x,y)
$-\frac{\pi}{4}$	$(2\sqrt{2},-3)$
0	$(2,0)$
$\frac{\pi}{4}$	$(2\sqrt{2},3)$
$\frac{3\pi}{4}$	$(-2\sqrt{2},-3)$
π	$(-2,0)$
$\frac{5\pi}{4}$	$(-2\sqrt{2},3)$



box from $a = 2$, $b = 3$; asymptotes through the corners; both branches run bottom to top

$\sec t = \frac{x}{2}, \quad \tan t = \frac{y}{3}$

$1 + \tan^2 t = \sec^2 t \Rightarrow 1 + \frac{y^2}{9} = \frac{x^2}{4}$ *identity*

$\boxed{\frac{x^2}{4} - \frac{y^2}{9} = 1}$ *t-chart: $(2,0)$ and $(-2,0)$, both branches: whole graph, done*

also right: $y = \pm 3\sqrt{\frac{x^2}{4} - 1}$ *keep $\pm$; $+\sqrt{}$ alone is wrong*

5. $y = -3(x+2)^2 + 5$

$x = t,\ y = -3(t+2)^2 + 5$ *replace x with t*

$\boxed{x = t,\ y = -3(t+2)^2 + 5,\ t \text{ in } (-\infty,\infty)}$

$x = t - 2:\ y = -3(t-2+2)^2 + 5$ *pick $t-2$ so the inside becomes t*

$\boxed{x = t - 2,\ y = -3t^2 + 5,\ t \text{ in } (-\infty,\infty)}$

Give both $x =$ and $y =$ ($t = x+2$ is not parametric form). "For t in $(-\infty,\infty)$" is assumed for a whole parabola; writing it is fine. Don't expand $-3(t+2)^2+5$; it's not asked and it wastes time.

6. $\overset{1}{\frac{1}{3}},\ \overset{2}{-\frac{2}{9}},\ \overset{3}{\frac{3}{27}},\ \overset{4}{-\frac{4}{81}}$

top $= n$, bottom $= 3^n$

signs $+,-,+,-\ \Rightarrow (-1)^{n+1}$ *first term positive, so $n+1$ ($(-1)^n$ makes it negative)*

$\boxed{a_n = (-1)^{n+1}\frac{n}{3^n}}$ *braces { } also fine*